

MLA0201-FUNDAMENTALS OF MACHINE LEARNING

APPLICATION BASED QUESTIONS

CO1-ASSESSMENT TOOL -1

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Q1) Implement The Concept Of Probability Distribution To Predict The Likelihood Of Spam Emails.

Aim

To implement the concept of probability distribution using the Naive Bayes algorithm for classifying emails as spam or non-spam.

Theory

Probability Distribution represents the likelihood of occurrence of different outcomes. In spam email detection, Naive Bayes uses Bayes' theorem to calculate the probability that an email belongs to the spam or ham category based on the occurrence of words. It assumes conditional independence among features, making it an efficient classifier for text classification.

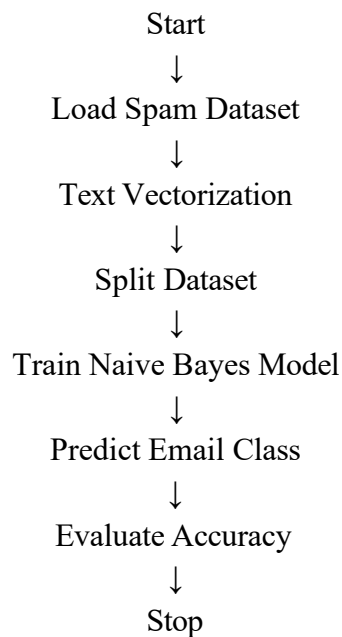
Dataset Used

- **Dataset Name:** Spam Email Dataset (spam.csv)
- **Source:** UCI Machine Learning Repository / Kaggle
- **Description:** Contains email messages labeled as Spam or Ham (Not Spam). The text is converted into numerical features using Count Vectorization.
- **Target Variable:** Label (Spam/Ham)

Algorithm

1. Import the required libraries.
2. Load the spam dataset.
3. Convert email text into numerical vectors.
4. Split the dataset into training and testing sets.
5. Train the Naive Bayes classifier.
6. Predict the class labels.
7. Calculate the accuracy.
8. Predict a new email.

Flowchart



Python Program

```
# =====  
# Spam Email Prediction using Naive Bayes  
# =====  
  
# Import Libraries  
import pandas as pd  
from sklearn.model_selection import train_test_split  
from sklearn.feature_extraction.text import CountVectorizer  
from sklearn.naive_bayes import MultinomialNB  
from sklearn.metrics import accuracy_score, classification_report  
  
# Load Dataset  
data = pd.read_csv("spam.csv", encoding="latin-1")  
  
# Display first five rows  
print("First 5 Rows:")  
print(data.head())  
  
# Keep only required columns  
data = data[['v1', 'v2']]  
  
# Rename columns  
data.columns = ['Label', 'Message']  
  
# Convert labels into numbers
```

```

data['Label'] = data['Label'].map( {'ham':0, 'spam':1} )
# Features and Target
X = data['Message']
y = data['Label']
# Convert text into numerical vectors
vectorizer = CountVectorizer()
X = vectorizer.fit_transform(X)
# Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
# Train Naive Bayes Model
model = MultinomialNB()
model.fit(X_train, y_train)
# Predict
prediction = model.predict(X_test)
# Accuracy
print("\nAccuracy:", accuracy_score(y_test, prediction))
print("\nClassification Report")
print(classification_report(y_test, prediction))
# Predict New Email
new_email = [
    "Congratulations! You have won a free iPhone. Click here to claim your reward."
]
new_email_vector = vectorizer.transform(new_email)
result = model.predict(new_email_vector)
print("\nNew Email Prediction:")
if result[0] == 1:
    print("Spam Email")
else:
    print("Not Spam")

```

Output

```
===== RESTART: C:/Users/shaik/OneDrive/Desktop/MLA0201/exp_email_spam.py =====
First 5 Rows:
   vl  ... Unnamed: 4
0  ham  ...      NaN
1  ham  ...      NaN
2 spam  ...      NaN
3  ham  ...      NaN
4  ham  ...      NaN

[5 rows x 5 columns]

Accuracy: 0.97847533632287

Classification Report
              precision    recall  f1-score   support

         0           0.99         0.99         0.99         965
         1           0.91         0.93         0.92         150

 accuracy                   0.98         1115
 macro avg           0.95         0.96         0.95         1115
weighted avg           0.98         0.98         0.98         1115

New Email Prediction:
Spam Email
```

Result

The Naive Bayes classifier successfully predicted whether an email was spam or non-spam using probability distribution concepts.

Q3) Design the steps of a learning system for student performance prediction.

Aim

To design and implement a machine learning system that predicts student performance using academic attributes.

Theory

A learning system extracts patterns from historical data and uses them to make predictions. In this experiment, a Decision Tree classifier learns from students' academic information such as study time, attendance, and previous grades to predict their performance.

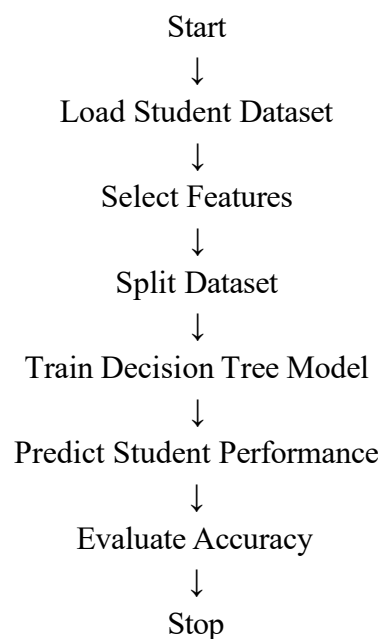
Dataset Used

- **Dataset Name:** Student Performance Dataset (student-mat.csv)
- **Source:** UCI Machine Learning Repository
- **Description:** Contains academic records of students including study time, failures, absences, and grades.
- **Target Variable:** Final Grade (G3) or Pass/Fail.

Algorithm

1. Import required libraries.
2. Load the student dataset.
3. Select input features and target.
4. Split the dataset.
5. Train the Decision Tree classifier.
6. Predict student performance.
7. Evaluate the model accuracy.

Flowchart



Python Program

```
# =====  
# Student Performance Prediction  
# =====  
  
# Import Libraries  
import pandas as pd  
from sklearn.model_selection import train_test_split  
from sklearn.tree import DecisionTreeClassifier  
from sklearn.metrics import accuracy_score, classification_report  
  
# Load Dataset  
data = pd.read_csv("student_data.csv", sep=",")  
  
# Display first five rows
```

```

print("First 5 Rows:")
print(data.head())
# Select Features
X = data[['studytime', 'failures', 'absences', 'G1', 'G2']]
# Target Variable
# G3 >= 10 --> Good Performance
# G3 < 10 --> Poor Performance
y = (data['G3'] >= 10).astype(int)
# Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
# Train Decision Tree Model
model = DecisionTreeClassifier(random_state=42)
model.fit(X_train, y_train)
# Predict
prediction = model.predict(X_test)
# Accuracy
print("\nAccuracy:", accuracy_score(y_test, prediction))
print("\nClassification Report")
print(classification_report(y_test, prediction))
# Predict New Student
new_student = pd.DataFrame({
    'studytime':[3],
    'failures':[0],
    'absences':[2],
    'G1':[14],
    'G2':[15]
})
result = model.predict(new_student)
print("\nNew Student Prediction:")

```

```

if result[0] == 1:
    print("Good Performance")
else:
    print("Poor Performance")

```

Output

```

===== RESTART: C:\Users\shaik\OneDrive\Desktop\MLA0201\exp Student Performance.py =====
First 5 Rows:
  school sex  age address famsize Pstatus  ...  Walc  health  absences  G1  G2  G3
0    GP   F   18     U    GT3      A  ...    1     3      6  5  6  6
1    GP   F   17     U    GT3      T  ...    1     3      4  5  5  6
2    GP   F   15     U    LE3      T  ...    3     3     10  7  8 10
3    GP   F   15     U    GT3      T  ...    1     5      2 15 14 15
4    GP   F   16     U    GT3      T  ...    2     5      4  6 10 10

[5 rows x 33 columns]

Accuracy: 0.8987341772151899

Classification Report
      precision    recall  f1-score   support

      0       0.85      0.85      0.85        27
      1       0.92      0.92      0.92        52

   accuracy       0.89
  macro avg       0.89
weighted avg       0.90

```

New Student Prediction:
Good Performance

Result

The Decision Tree classifier successfully predicted student performance based on academic features.

Q4) Develop a weather forecasting model using random variables and probability.

Aim

To develop a weather forecasting model using Gaussian Naive Bayes based on probability concepts.

Theory

Weather forecasting predicts future weather conditions using atmospheric measurements such as temperature, humidity, wind speed, and pressure. Gaussian Naive Bayes assumes that continuous features follow a Gaussian distribution and estimates the probability of each weather condition.

Dataset Used

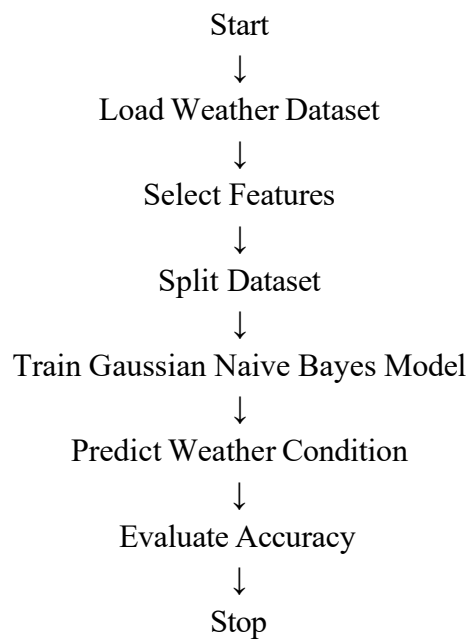
- **Dataset Name:** Weather History Dataset (weatherHistory.csv)
- **Source:** Kaggle
- **Description:** Contains historical weather observations including temperature, humidity, pressure, wind speed, visibility, and weather summaries.

- **Features Used:** Temperature, Humidity, Wind Speed, Pressure
- **Target Variable:** Summary

Algorithm

1. Import required libraries.
2. Load the weather dataset.
3. Select weather features.
4. Split the dataset.
5. Train the Gaussian Naive Bayes classifier.
6. Predict weather conditions.
7. Evaluate the model.

Flowchart



Python Program

```

# =====
# Weather Forecasting using Probability Theory
# =====
# Import Libraries

```



```

import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, classification_report
from sklearn.preprocessing import LabelEncoder

# Load Dataset
data = pd.read_csv("weatherHistory.csv")
print("First Five Rows")
print(data.head())

# Select Features
X = data[['Temperature (C)',
          'Humidity',
          'Wind Speed (km/h)',
          'Pressure (millibars)']]

# Target Variable
y = data['Summary']

# Convert Text Labels into Numbers
encoder = LabelEncoder()
y = encoder.fit_transform(y)

# Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

# Train Model
model = GaussianNB()
model.fit(X_train, y_train)

# Prediction
prediction = model.predict(X_test)

# Accuracy
print("\nAccuracy:", accuracy_score(y_test, prediction))
print("\nClassification Report")

```

```

print(classification_report(y_test, prediction))

# Predict New Weather

new_weather = pd.DataFrame({
    'Temperature (C)': [28],
    'Humidity': [0.75],
    'Wind Speed (km/h)': [12],
    'Pressure (millibars)': [1015]
})

result = model.predict(new_weather)

print("\nPredicted Weather Condition:")

print(encoder.inverse_transform(result))

```

Output

```

=== RESTART: C:\Users\shaik\OneDrive\Desktop\MLA0201\exp Weather forecast.py ===

```

```

First Five Rows

```

	Formatted Date	...	Daily Summary
0	2006-04-01 00:00:00.000 +0200	...	Partly cloudy throughout the day.
1	2006-04-01 01:00:00.000 +0200	...	Partly cloudy throughout the day.
2	2006-04-01 02:00:00.000 +0200	...	Partly cloudy throughout the day.
3	2006-04-01 03:00:00.000 +0200	...	Partly cloudy throughout the day.
4	2006-04-01 04:00:00.000 +0200	...	Partly cloudy throughout the day.

```

[5 rows x 12 columns]

```

```

Accuracy: 0.4238764190555181

```

```

Classification Report

```

	precision	recall	f1-score	support
0	0.25	0.42	0.31	12
1	0.00	0.00	0.00	1
2	0.27	0.75	0.40	4
3	0.37	0.28	0.32	94
4	0.42	0.68	0.52	94
5	0.41	0.22	0.29	67
6	0.62	0.06	0.11	2183
8	0.00	0.00	0.00	8
9	0.00	0.00	0.00	3
10	0.00	0.00	0.00	3
11	0.04	0.41	0.07	17
12	0.40	0.78	0.53	1446
13	0.07	0.15	0.09	13
14	0.00	0.00	0.00	1
15	0.50	0.25	0.33	4
16	0.00	0.00	0.00	16
17	0.37	0.35	0.36	5618
18	0.36	0.34	0.35	3284
19	0.51	0.58	0.54	6392
20	0.00	0.00	0.00	0
21	0.00	0.00	0.00	0
23	0.00	0.00	0.00	1
24	0.30	0.75	0.43	8
25	0.20	0.67	0.31	6
26	0.67	0.12	0.21	16
accuracy			0.42	19291
macro avg	0.23	0.27	0.21	19291
weighted avg	0.45	0.42	0.40	19291

```
Predicted Weather Condition:
['Partly Cloudy']
|
```

Result

The Gaussian Naive Bayes classifier successfully predicted weather conditions using probability concepts.

Q6) Apply Conditional Probability concepts to design a fraud detection model for suspicious banking transactions.

Aim

To develop a fraud detection model using conditional probability and the Naive Bayes classifier.

Theory

Conditional Probability measures the probability of an event occurring given that another event has occurred. In fraud detection, it is used to estimate the likelihood that a transaction is fraudulent based on its transaction characteristics. Naive Bayes applies Bayes' theorem to classify transactions efficiently.

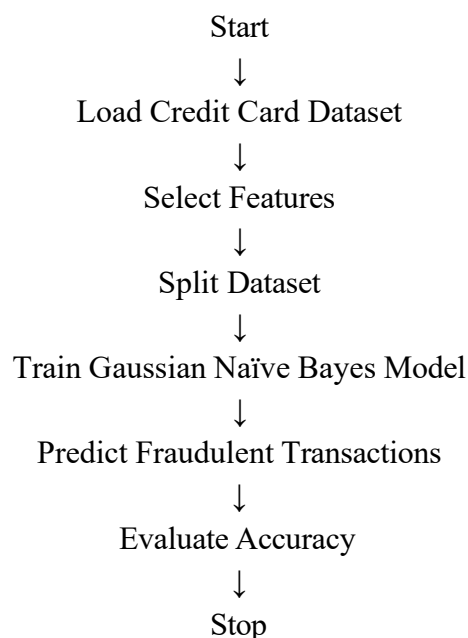
Dataset Used

- **Dataset Name:** Credit Card Fraud Detection Dataset (creditcard.csv)
- **Source:** Kaggle
- **Description:** Contains credit card transactions made by European cardholders. Features V1 to V28 are anonymized for privacy, along with Time and Amount.
- **Target Variable:** Class
 - 0 – Genuine Transaction
 - 1 – Fraudulent Transaction

Algorithm

1. Import required libraries.
2. Load the credit card dataset.
3. Separate features and target.
4. Split the dataset.
5. Train the Gaussian Naïve Bayes classifier.
6. Predict fraudulent transactions.
7. Evaluate the model performance.

Flowchart



Python Program

```

# =====
# Credit Card Fraud Detection using
# Conditional Probability (Naive Bayes)
# =====

# Import Libraries
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import accuracy_score, classification_report

# Load Dataset
data = pd.read_csv("creditcard.csv")

print("First Five Rows")
print(data.head())

# Select Features
X = data.drop('Class', axis=1)

# Target Variable
# 0 = Genuine
# 1 = Fraud
y = data['Class']

# Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

# Train Model
model = GaussianNB()
model.fit(X_train, y_train)

# Prediction
prediction = model.predict(X_test)

# Accuracy
print("\nAccuracy:", accuracy_score(y_test, prediction))

```

```
print("\nClassification Report")
print(classification_report(y_test, prediction))
# Predict New Transaction
new_transaction = pd.DataFrame({
    'Time':[50000],
    'V1':[0.25],
    'V2':[1.20],
    'V3':[0.50],
    'V4':[1.10],
    'V5':[0.30],
    'V6':[-0.10],
    'V7':[0.20],
    'V8':[0.05],
    'V9':[0.60],
    'V10':[0.15],
    'V11':[0.40],
    'V12':[0.80],
    'V13':[0.20],
    'V14':[-0.50],
    'V15':[0.10],
    'V16':[0.15],
    'V17':[0.05],
    'V18':[0.10],
    'V19':[0.30],
    'V20':[0.15],
    'V21':[0.05],
    'V22':[0.12],
    'V23':[0.02],
    'V24':[0.10],
    'V25':[0.20],
    'V26':[0.15],
    'V27':[0.01],
    'V28':[0.02],
    'Amount':[250]
```

```

})

result = model.predict(new_transaction)

print("\nTransaction Prediction:")

if result[0] == 1:

    print("Fraudulent Transaction")

else:

    print("Genuine Transaction")

```

Output

```

===== RESTART: C:\Users\shaik\OneDrive\Desktop\MLA0201\exp credit card.py ==
First Five Rows
Time      V1      V2      V3      ...      V27      V28      Amount  Class
0    0.0 -1.359807 -0.072781 2.536347 ... 0.133558 -0.021053 149.62    0
1    0.0  1.191857  0.266151 0.166480 ... -0.008983  0.014724   2.69    0
2    1.0 -1.358354 -1.340163 1.773209 ... -0.055353 -0.059752 378.66    0
3    1.0 -0.966272 -0.185226 1.792993 ...  0.062723  0.061458 123.50    0
4    2.0 -1.158233  0.877737 1.548718 ...  0.219422  0.215153  69.99    0

[5 rows x 31 columns]

Accuracy: 0.9930128857835048

Classification Report
      precision    recall  f1-score   support

     0       1.00      0.99      1.00     56864
     1       0.15      0.63      0.24        98

 accuracy
macro avg      0.57      0.81      0.62     56962
weighted avg      1.00      0.99      1.00     56962

Transaction Prediction:
Genuine Transaction
|

```

Result

The Gaussian Naive Bayes classifier successfully identified genuine and fraudulent banking transactions using conditional probability.

Q7) Illustrate how Bayes Decision Theory can be applied in medical diagnosis for disease prediction.

Aim

To apply Bayes Decision Theory for predicting heart disease using patient medical information.

Theory

Bayes Decision Theory is a probabilistic approach that selects the class with the highest posterior probability. In healthcare, it helps predict whether a patient is likely to have heart disease based on clinical measurements such as age, blood pressure, cholesterol level, heart rate, and ECG results.

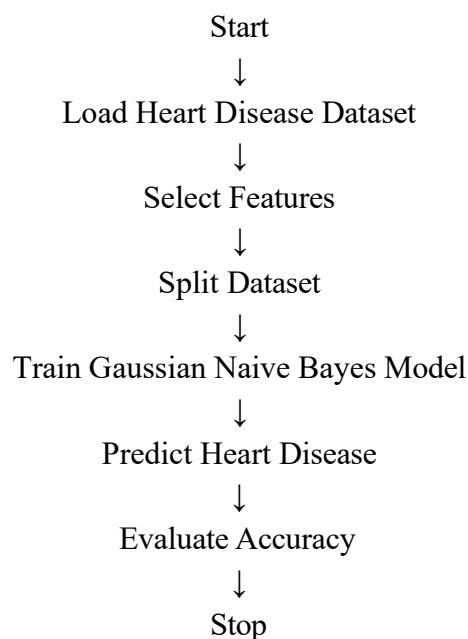
Dataset Used

- **Dataset Name:** Heart Disease Dataset (heart.csv)
- **Source:** UCI Machine Learning Repository / Kaggle
- **Description:** Contains clinical information of patients used for predicting the presence or absence of heart disease.
- **Features:** Age, Sex, Chest Pain Type, Resting Blood Pressure, Cholesterol, Fasting Blood Sugar, Resting ECG, Maximum Heart Rate, Exercise-Induced Angina, ST Depression, Slope, Number of Major Vessels, Thalassemia.
- **Target Variable:** Target
 - 0 – No Heart Disease
 - 1 – Heart Disease

Algorithm

1. Import required libraries.
2. Load the heart disease dataset.
3. Separate features and target.
4. Split the dataset.
5. Train the Gaussian Naive Bayes classifier.
6. Predict heart disease.
7. Evaluate the classifier.
8. Predict disease for a new patient.

Flowchart



Python Program

```
# Bayes Decision Theory for Medical Diagnosis

import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import accuracy_score, classification_report

# Load dataset

df = pd.read_csv("heart.csv")

# Display first five rows

print("First Five Rows")

print(df.head())

# Features and Target

X = df.drop("target", axis=1)

y = df["target"]

# Split dataset

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Train Bayes Classifier

model = GaussianNB()

model.fit(X_train, y_train)

# Prediction

y_pred = model.predict(X_test)

# Accuracy

print("\nAccuracy:", accuracy_score(y_test, y_pred))

# Classification Report

print("\nClassification Report\n")

print(classification_report(y_test, y_pred, zero_division=0))

# Predict new patient

new_patient = pd.DataFrame({
    "age": [63],
    "sex": [1],
    "cp": [3],
    "trestbps": [145],
```

```

        "chol":[233],
        "fbs":[1],
        "restecg":[0],
        "thalach":[150],
        "exang":[0],
        "oldpeak":[2.3],
        "slope":[0],
        "ca":[0],
        "thal":[1]
    })
    prediction = model.predict(new_patient)
    print("\nMedical Diagnosis:")
    if prediction[0] == 1:
        print("Heart Disease Detected")
    else:
        print("No Heart Disease")

```

Output

```

===== RESTART: C:\Users\shaik\OneDrive\Desktop\MLA0201\exp medica
First Five Rows
  age  sex  cp  trestbps  chol  fbs  ...  exang  oldpeak  slope  ca  thal  target
0   52   1   0    125    212  0  ...    0      1.0      2   2    3      0
1   53   1   0    140    203  1  ...    1      3.1      0   0    3      0
2   70   1   0    145    174  0  ...    1      2.6      0   0    3      0
3   61   1   0    148    203  0  ...    0      0.0      2   1    3      0
4   62   0   0    138    294  1  ...    0      1.9      1   3    2      0

[5 rows x 14 columns]
Accuracy: 0.8
Classification Report

```

	precision	recall	f1-score	support
0	0.87	0.71	0.78	102
1	0.75	0.89	0.82	103
accuracy			0.80	205
macro avg	0.81	0.80	0.80	205
weighted avg	0.81	0.80	0.80	205

```

Medical Diagnosis:
No Heart Disease

```

Result

The Gaussian Naive Bayes classifier successfully predicted the presence or absence of heart disease using Bayes Decision Theory.
